WORKED SOLUTION

STEP III 1989, Question 15

Probability and Statistics • Distribution of the Median

STEP III 1989, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q15

The key idea

The median of $2n+1$ values lies close to z exactly when one value falls in a tiny interval at z, n values fall below it and n above it. Parts (i) and (ii) give the probabilities of the "all below" and "all above" pieces, and counting the ways to choose which values play which role gives the density. After that, the fact that a density integrates to 1 evaluates the integral for free, and the same trick, used with $n+1$ in place of n, gives the variance.

Solution

Throughout, X has density $\frac{1}{2c}$ on $[-c, c]$, so for $-c \le k \le c$

$$\mathrm{P}(X > k) = \frac{c-k}{2c}, \qquad \mathrm{P}(X < k) = \frac{c+k}{2c}.$$

Part (i)

The n values are independent, so we multiply:

$$\mathrm{P}(\text{all } n \text{ values exceed } k) = \left(\frac{c-k}{2c}\right)^n.$$

Part (ii)

In the same way,

$$\mathrm{P}(\text{all } n \text{ values are less than } k) = \left(\frac{c+k}{2c}\right)^n.$$

The density of the median

Let $X_1, \ldots, X_{2n+1}$ be the sample, and Z its median, the $(n+1)$th smallest value. Since X is continuous, ties happen with probability 0, so we ignore them. Every X_i lies in

$[-c, c]$, so Z does too.

Take z in $(-c, c)$ and a small $\delta z > 0$. Up to terms of order $(\delta z)^2$, the event $z < Z \leq z + \delta z$ happens exactly when

- one of the values lies in $(z, z + \delta z]$,
- n of the other values are less than z, and
- the remaining n values are greater than $z + \delta z$.

(Arrangements with two or more values in the little interval have probability of order $(\delta z)^2$, which doesn't matter in the limit.)

Ask yourself How many ways are there to assign the $2n + 1$ values to these three roles?

There are $2n + 1$ choices for the value in the little interval, and then $\binom{2n}{n}$ ways to choose which n of the rest go below z. That gives

$$(2n+1)\binom{2n}{n} = \frac{(2n+1)!}{(n!)^2}$$

assignments. They are mutually exclusive, and each has the same probability. For a fixed assignment, independence and parts (i) and (ii) with $k = z$ give the probability

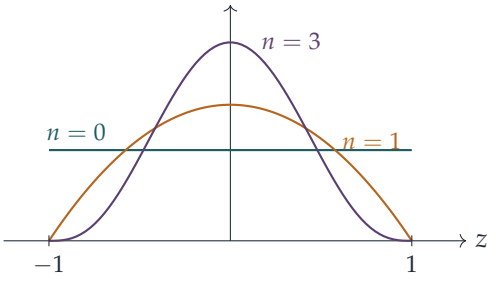
$$\frac{\delta z}{2c}\left(\frac{c+z}{2c}\right)^n \left(\frac{c-z-\delta z}{2c}\right)^n.$$

Adding over the assignments, dividing by δz and letting $\delta z \to 0$, the density of Z is

$$\begin{aligned} g(z) &= \frac{(2n+1)!}{(n!)^2} \cdot \frac{1}{2c} \cdot \frac{(c+z)^n (c-z)^n}{(2c)^{2n}} \\ &= \frac{(2n+1)!}{(n!)^2 (2c)^{2n+1}} (c^2 - z^2)^n, \qquad -c \leq z \leq c, \end{aligned}$$

using $(c + z)(c - z) = c^2 - z^2$. This is the required result.

The picture below shows the density for $c = 1$. When $n = 0$ the "median" is a single value, which is uniform. As n grows, the median concentrates near 0.



The integral

A probability density integrates to 1 over its range, so

$$\frac{(2n+1)!}{(n!)^2(2c)^{2n+1}}\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = 1,$$

and therefore

$$\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = \frac{(n!)^2(2c)^{2n+1}}{(2n+1)!}.$$

This holds for every integer $n \geq 0$, and we use that in a moment.

Mean and variance

The density g is an even function of z, so $zg(z)$ is odd and its integral over the symmetric interval $[-c, c]$ is zero:

$$\mathrm{E}(Z) = 0.$$

Hence $\mathrm{var}(Z) = \mathrm{E}(Z^2)$. Write $K_n = \dfrac{(2n+1)!}{(n!)^2(2c)^{2n+1}}$ and $I_n = \displaystyle\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = \frac{1}{K_n}$.

? **Ask yourself** How can we integrate $z^2(c^2-z^2)^n$ without expanding it?

Write $z^2 = c^2 - (c^2 - z^2)$. Then

$$\begin{aligned}\mathrm{E}(Z^2) &= K_n\int_{-c}^{c}\big(c^2-(c^2-z^2)\big)(c^2-z^2)^n\,\mathrm{d}z\\ &= K_n\big(c^2 I_n - I_{n+1}\big) = c^2 - \frac{K_n}{K_{n+1}},\end{aligned}$$

using $K_n I_n = 1$ and $I_{n+1} = 1/K_{n+1}$, which is the integral result with $n+1$ in place of n. Now

$$\frac{K_n}{K_{n+1}} = \frac{(2n+1)!}{(2n+3)!}\cdot\frac{((n+1)!)^2}{(n!)^2}\cdot\frac{(2c)^{2n+3}}{(2c)^{2n+1}} = \frac{(n+1)^2\cdot 4c^2}{(2n+2)(2n+3)} = \frac{2(n+1)c^2}{2n+3}.$$

Therefore

$$\mathrm{var}(Z) = c^2\left(1-\frac{2n+2}{2n+3}\right) = \frac{c^2}{2n+3}.$$

Checks

- When $n = 0$ the median is just one value of X. The density becomes $\frac{1}{2c}$, the integral is $2c$, and the variance is $\frac{c^2}{3}$, which is the variance of the uniform distribution on $[-c, c]$.
- When $n = 1$ and $c = 1$, the integral formula gives $\int_{-1}^{1}(1-z^2)\,\mathrm{d}z = \frac{1\cdot 8}{6} = \frac{4}{3}$, which is correct.
- A simulation of 10^5 samples of size 5 ($n = 2$) with $c = 1.7$ gave a median variance of 0.415, against $c^2/7 \approx 0.413$.

Common slips

- Forgetting the factor $2n+1$ for the choice of the middle value, and using only $\binom{2n}{n}$.
- Missing the factor $\frac{1}{2c}$ for the middle value's own density, which leaves the power $(2c)^{2n}$ instead of $(2c)^{2n+1}$.
- Expanding $(c^2 - z^2)^n$ binomially to find $\mathrm{E}(Z^2)$. It can be done, but the sum is hard to simplify.

Beyond STEP

The median is one of the *order statistics* of a sample. The same counting argument gives the density of the rth smallest of m values from any continuous distribution, as $\frac{m!}{(r-1)!\,(m-r)!} F^{r-1}(1-F)^{m-r} f$. For the uniform distribution these are *beta distributions*, and the integral you deduced is a case of the beta function $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$. The variance $c^2/(2n+3)$ shrinks like $1/n$, which is the first step towards the asymptotic normality of the sample median. Blitzstein and Hwang's *Introduction to Probability* covers order statistics and the beta distribution well.

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